Lib-DS

Reg. No. | | | | | | | | | | |

[18MTDS101]

FIRST SEMESTER M.TECH EXAMINATION, JANUARY 2020

DISCRETE MATHEMATICS AND LINEAR ALGEBRA

Time: 03 Hours **Maximum Marks:100**

Instructions :
- **i.** Answer Five full questions. Question No. 1 and 2 are Compulsory.
- **ii.** Draw neat sketches wherever necessary
- **iii.** Missing data may be assumed suitably.

1. a. Define set and types of sets with a suitable example. **07**

b. State whether the argument given is valid or not. If it is valid, identify the tautology or tautologies on which it is based. **06**

If I try hard and I have talent, then I will become musician.

If I become a musician, then I will be happy.

Therefore, If I will not be happy, then I did not try hard or I do not have talent.

c. i. Suppose the universe consists of all integers. Consider the following open statements **07**

$p(x): x \le 3$, $q(x): x+1$ is odd. write down the truth values of the following

i. $p(0) \to q(0)$

ii. $p(2) \vee q(0)$

iii. $p(4) \wedge q(3)$

ii. Negate and simplify each of the following

i. $\exists x, [p(x) \vee q(x)]$

ii. $\forall x, [p(x) \wedge \neg q(x)]$

iii. $\forall x, [p(x) \to q(x)]$

iv. $\exists x, [\{p(x) \vee q(x)\} \to r(x)]$

2. a. How many arrangements are there for all letters in the word SOCIOLOGICAL? **07**

i. In how many of these arrangements, A and G are adjacent?

ii. All the vowels are adjacent

b. In how many ways can the 26 letters of the alphabet by permuted so that none of the patterns car, dog, pun, or byte occurs? **07**

c. Solve the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$ *for* $n \ge 2$ with initial condition $a_1 = 5, a_2 = 3$. **06**

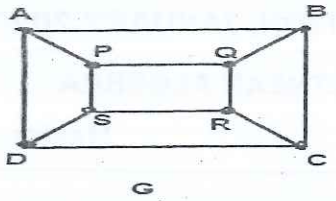
3. a. Define the following terms with an example for each **06**

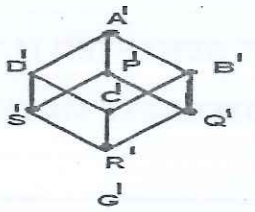
i. Graph.

ii. Spanning sub-graph.

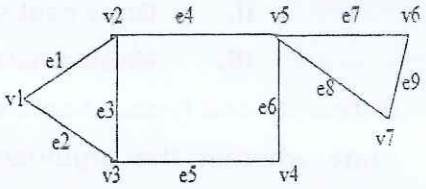
iii. Connected graph.

b. Define isomorphic graph and prove that following graphs are isomorphic. **07**





c. For the given graph, determine **07**



i. A walk from v2 to v4 which is not a trail.

ii. v2-v4 trail which is not a path.

iii. A path from v2 to v4.

iv. A closed walk from v2 to v2 which is not a circuit.

v. A circuit from v2 to v2 which is not a cycle.

vi. A cycle from v2 to v2.

OR

4 a. Define the following terms with an example for each **06**

i. Bipartite graph.

ii. Euler trail.

iii. Rooted tree.

b. Define chromatic polynomial. Find chromatic polynomial P_G for the given graph using decomposition theorem and also find chromatic number X(G) **07**

c. If A={1,2,3,4} and R={(1,2)(2,3)(3,4)(2,1)}.Find the transitive closure R^{∞} by using Warshall's Algorithm . **07**

5. a. Investigate the values of λ and μ so that the equations **07**

$$x+y+z=6, x+2y+3z=10, x+2y+\lambda z=\mu$$

have

i. a unique solution.

ii. an infinite number of solution.

iii. no solution.

b. Employing the Gauss Siedel method to solve the system of equations $x+y+54z=110$, $27x+6y-z=85$ and $6x+15y+2z=72$. **07**

c. The following is a description of a two-industry model: we have two industries, electric company E and water company W. Output for both companies is measured in dollars. The electric company uses both electricity and water (input) in the production of electricity (output), and the water company uses both electricity and water (input) in the production of water (output). Suppose that the production of each dollar's worth of electricity requires \$0.3 worth of electricity and \$0.1 worth **06**

of water, and the production of each dollar's worth of water requires \$0.2 worth of electricity and \$0.4 worth of water. The final demand from the outside sector of the economy is \$12 million for electricity and \$8 million for water. Justify that

i. How much electricity and water should be produced to meet this final demand?

ii. How can we find the internal demand of E and W?

OR

6. a. Show that $\{u_1, u_2, u_3\}$ is an orthogonal set where $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$ **07**

Express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as linear combination of vector S.

b. Apply the Gram–Schmidt orthonormalization process to the basis $B = \{(1,1,1,1), (0,1,1,1), (0,0,1,1)\}$ of the inner product space R^4 to find an orthogonal basis for R^4. **07**

c. Find the Least square solutions to the system $AX = B$ where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$ **06**

7. a. Verify Cayley-Hamilton Theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and find its inverse. **05**

b. Diagonalize the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ and also find A^4. **07**

c. Solve the system $X' = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} X + \begin{bmatrix} 3e^t \\ e^t \end{bmatrix}$ and also discuss the stability of the system. **08**

OR

8. a. If $q(x) = 8x^2 + 7y^2 + 3z^2 - 12xy + 4xz - 8yz$ find the rank, index, signature and nature of quadratic form. **05**

b. Verify that $T: V_3(R) \to V_3(R)$ defined by $T(x, y, z) = (3x - 2y + z, x - 3y - 2z)$ is linear transformation. **07**

c. Diagonalization of symmetric matrix A= $\begin{bmatrix} 16 & -4 \\ -4 & 1 \end{bmatrix}$ **08**